

PING-YU YANG

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EDUCATION

National Taiwan University	Aug 2024 – Jun 2026 (Expected)
Master's degree in Computer Science and Information Engineering	
Research: Semantic Communication, Deep Learning, Wireless Communication	
National Yang Ming Chiao Tung University	Sep 2020 – Jun 2024
Bachelor's degree in Computer Science	
Overall GPA 4.04/4.3 GPA of CS courses 4.17/4.3	

EXPERIENCES

Taiwan Semiconductor Manufacturing Company, Hsinchu, Taiwan	IT Intern	Jul 2024 - Aug 2024
- Built the frontend for the internal leave-approval platform using React and optimized a core Java backend API by redesigning the data-access algorithm from $O(n^2)$ to $O(n)$. - Achieved over 1000x speedup in the production environment, directly enhancing system scalability and reliability for a platform serving 20,000+ employees.		
BingMap Direction team, Microsoft, Taiwan	Research and Development Intern	Jul 2023 – Jun 2024
- Reduced map quality debugging time for the BingMap team by developing and deploying a full-stack internal routing debugger. - Optimized C# backend APIs to automate complex data retrieval, reducing redundant operational steps and improving system response time.		
Logitech, Hsinchu, Taiwan	Software Engineer Intern	Mar 2023 - Jun 2023
- Built a robust Python/Flask backend that automated and scaled data parsing pipelines from Excel inputs, replacing a highly manual data management process. - Deployed the automated system, reducing manual operations by over 60% and delivering data to an Angular visualization tool used by the engineering team.		

AWARDS

Silver medal 12th place, ICPC Asia Taipei Regional	2021
3rd prize award National Taiwan Collegiate Programming Contest	2021
2nd place National Yang Ming Chiao Tung Annual Programming Competition	2021
Academic Achievement Awards (Top 5%) Fall 2021 (GPA 4.25), Spring 2022(GPA 4.26)	

SKILLS

Programming Languages	C/C++, C#, Python, Java, Golang, TypeScript, SQL, Shell Script
Frameworks and Libraries	Vue3, React, Angular, Flask, PyTorch, Keras, Scikit-learn
Tools and Platforms	Git, Linux, Docker, System and network administration

PROJECTS

Generating Jazz Drum Comping With AI	Machine Learning, Tensorflow, Python	GitHub
Built a TensorFlow-based CNN/RNN pipeline to generate jazz drum comping, designing the MIDI preprocessing stage and sequence-model architecture from scratch.		
Meal Provider Platform	TypeScript, Vue3, SQL, Flask, Docker	GitHub
- Staff restaurant platform that store owner can add the new meal and user can order the combo. - Frontend: Designed and built a responsive frontend using Vue3 and TypeScript. - Backend: Use Flask and SQL to store the owner and user data and handle APIs.		